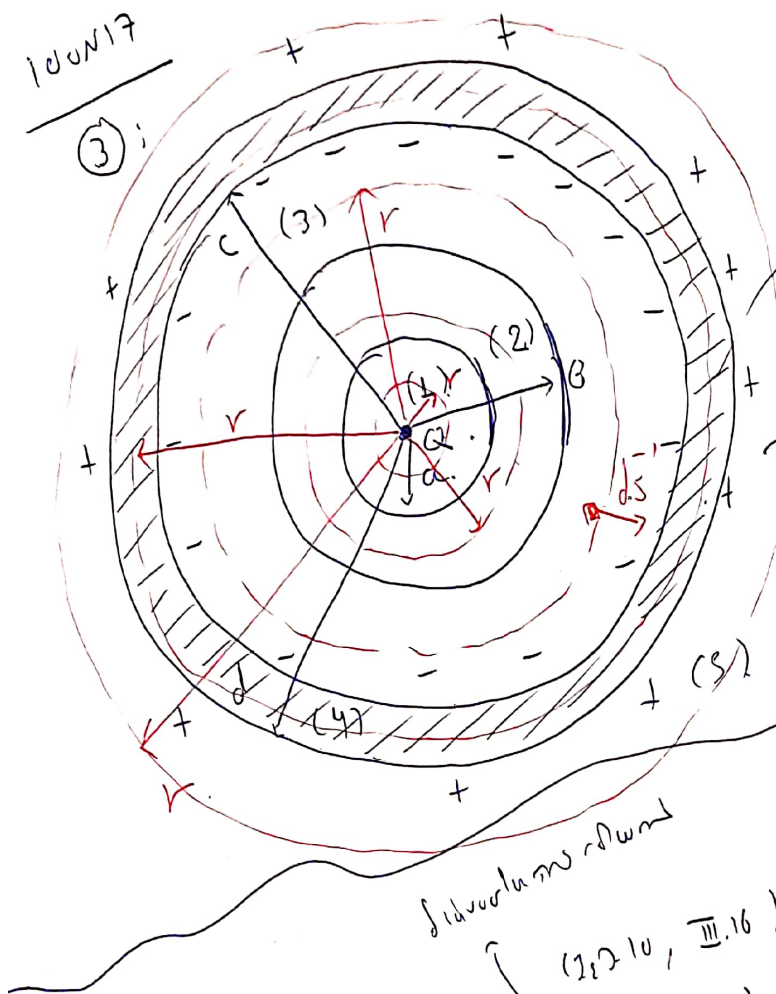


100N17
③;



- $\epsilon_1 = \epsilon_0 = \epsilon_3$.
 $\epsilon_2 = \epsilon_r \epsilon_0$.
 $\sigma_f \rightarrow \omega$.
 (α) $\vec{E}, \vec{D}, \vec{P} = ?$
 (β) $\rho_s|_c, \rho_s|_d = ?$
 (γ) φορτία ρωμάτων = ?

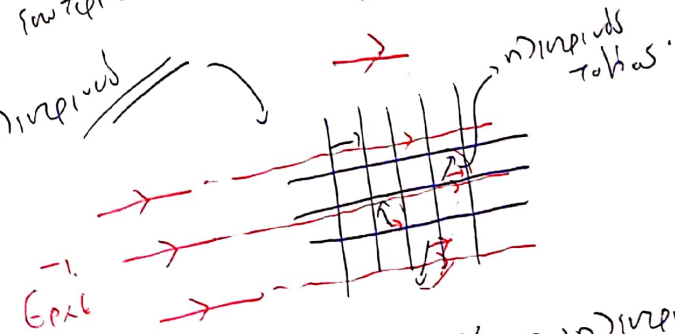
→ δώω n/r είδραμα
 για $r = c$ έχω $\epsilon \epsilon_{\text{ραγ}} = -Q$
 για $r = d$ έχω $\epsilon \epsilon_{\text{ραγ}} = +Q$.

(δίοτλια φορτία)
 Χωρίς φορτία
 Ενφωτισμένο φορτία
 ρωμάτων
 $\vec{P}_v = -\nabla P$ (C/m^3)
 $\vec{P} = \epsilon_0 (\vec{E} - \vec{E}_0)$
 $\vec{P} = \epsilon_0 (\vec{E} - \vec{E}_0)$ (C/m^2)

(21210, III.16)
 $\vec{P} = \chi_e \cdot \epsilon_0 \vec{E}$ (C/m^2)

δρα, για
 $\vec{D} = \epsilon_0 \vec{E} + \vec{P} = \epsilon_0 \vec{E} + \chi_e \cdot \epsilon_0 \vec{E} = \epsilon_0 (\chi_e + 1) \vec{E} = \epsilon_r \epsilon_0 \vec{E}$
 $\vec{D} = \epsilon_0 \vec{E}$
 για $\epsilon_r = 1, \epsilon = \epsilon_0$
 $\vec{P} = 0$

Zoom στο εσωτερικό
 ενός διηλεκτρικού



$\chi_e \sim$ διηλεκτρική
 $\chi_e = \epsilon_r - 1$ ή $\epsilon_r = 1 + \chi_e$

→ διηλεκτ. αν έχω ϵ_r έχω
 $\epsilon = \epsilon_0$, τότε $\chi_e = 0$
 ($\epsilon_r = 1$)

(1), (2), (3), (5):

$$D \cdot 4\pi r^2 = Q \rightarrow$$

$$\rightarrow \vec{D} = \frac{Q}{4\pi r^2} \hat{r} \quad (C/m^2)$$

$$\vec{E}_1 = \vec{E}_3 = \vec{E}_5 = \frac{Q}{4\pi \epsilon_0 r^2} \hat{r}, (V/m) \rightarrow \vec{P}_1 = \vec{0}$$

$$\vec{E}_2 = \frac{Q}{4\pi \epsilon_r \epsilon_0 r^2} \hat{r}, (V/m) \rightarrow \vec{P}_2 = (\epsilon_r - 1) \epsilon_0 \cdot \frac{Q}{4\pi \epsilon_r \epsilon_0 r^2} \hat{r} (C/m^2) \quad \vec{P}_4 = \vec{0}$$

Λόγω συμμετρίας
 χρησιμοποιούμε
 το θεώρημα Gauss
 $\vec{E} = |\vec{E}| \hat{r}, (V/m)$
 $\vec{D} = |\vec{D}| \hat{r}, (C/m^2)$

(4) $\oint \vec{D} \cdot d\vec{S} = Q_{enc} \rightarrow$
 in r, S

$$\rightarrow \int_0^a \int_0^{2\pi} \int_0^{2\pi} D \hat{r} \cdot r^2 \sin\theta d\theta d\phi dr = Q + \cancel{Q_{enc}}$$

$$D \cdot 4\pi r^2 = Q - Q = 0 \rightarrow$$

$$\rightarrow \vec{D}_4 = \vec{0}$$

$$\vec{E}_4 = \vec{0}$$

(β) $P_S|_{r=c} = n_{03 \rightarrow 4} (\vec{D}_3 - \vec{D}_4)|_{r=c} = \hat{r} \cdot \left(0 - \frac{Q}{4\pi r^2} \hat{r} \right) |_{r=c} = \frac{-Q}{4\pi c^2} (C/m^2)$ (Due to surface charge = $+Q$ at $r=c$)

$P_S|_{r=d} = n_{04 \rightarrow 5} (\vec{D}_5 - \vec{D}_4)|_{r=d} = \hat{r} \cdot \left(\frac{Q}{4\pi r^2} \hat{r} - 0 \right) |_{r=d} = \frac{Q}{4\pi d^2} (C/m^2)$ (surface charge for $r=c$)
 $r=d$

(γ) Επιφανειακή φορτία ρθωμε για $r=a$ σε $r=b$ μόνο στο χώρο $1 \leq r \leq 2$ για $a \leq r \leq b$

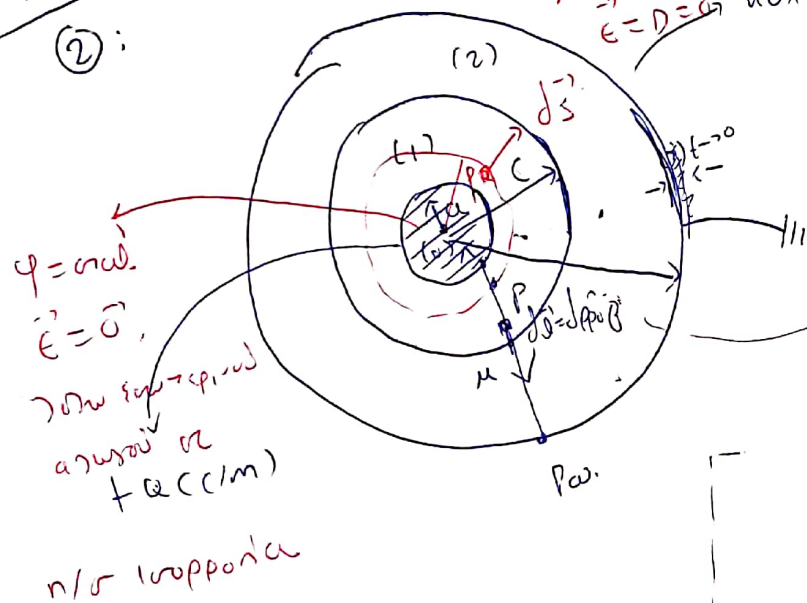
$$P_{sb}|_{r=a} = \cancel{n_{01} \cdot \vec{r}_0} \cdot (\cancel{\vec{r}_1 - \vec{r}_2})|_{r=a} = \cancel{\vec{r}_0} \cdot \left(- \frac{(\epsilon_{r2}-1)Q}{4\pi\epsilon_{r2}\epsilon_0} \frac{1}{r^2} \cancel{\vec{r}_0} \right)|_{r=a} = - \frac{Q(\epsilon_{r2}-1)}{4\pi\epsilon_{r2}\epsilon_0 a^2} \quad -3-$$

$$P_{sb}|_{r=b} = \cancel{n_{02} \cdot \vec{r}_0} (\cancel{\vec{r}_2 - \vec{r}_3})|_{r=b} = \cancel{\vec{r}_0} \cdot \left(\frac{(\epsilon_{r2}-1)Q}{4\pi\epsilon_{r2}\epsilon_0} \frac{1}{r^2} \cancel{\vec{r}_0} \right)|_{r=b} = \frac{Q(\epsilon_{r2}-1)}{4\pi\epsilon_{r2}\epsilon_0 b^2} \quad (C/m^2)$$

$$P_{vb} = -\nabla P_2 = - \frac{1}{r^2} \frac{\partial(r^2 P_2)}{\partial r} = - \frac{1}{r^2} \frac{\partial}{\partial r} \left(\frac{Q(\epsilon_{r2}-1)}{4\pi\epsilon_{r2}\epsilon_0 r^2} \right) = 0, \quad (C/m^3)$$

(I.12
α2.96)

$\varphi \in \mathcal{B}^{11}$
(2):



$\varphi = 0$
 $\vec{E} = \vec{D} = \vec{0}$
 - $\vec{E} = \vec{D} = \vec{0}$ inside the dielectric.
 - $\vec{E} = \vec{D} = \vec{0}$ outside the dielectric.

$$C = C_1/2, \quad \epsilon_1 = 2\epsilon_0, \quad \epsilon_2 = \left(3 - \frac{P}{C}\right)\epsilon_0, \quad (F/m)$$

$$\varphi_2(r) = \frac{k}{\epsilon_0} \left[\ln 2 + \ln \left(\frac{3C-P}{P} \right) \right], \quad (V), \quad k \in \mathbb{R}$$

and $\varphi_2(r) = 0$ at $r = b$.

(a) $\vec{E}_2, \vec{D}_2 = ?$

(b) $k = ?$ so that $P = C$ is a fixed value.

Na βρωσω τα P_S αναλ.

1
L (a) $\vec{E}_2 = -\nabla\varphi_2 = -\frac{\partial\varphi_2}{\partial\rho}\hat{\rho}_0 = -\frac{\kappa}{\epsilon_0}\left(0 - \frac{3c}{\rho(3c-\rho)}\right)\hat{\rho}_0 = \frac{3\kappa c}{\epsilon_0\rho(3c-\rho)}\hat{\rho}_0 \text{ (V/m)}$

Εφαρμοζοντας
στον φ_2
στα σημεία $\vec{E}_2, \vec{D}_2$ έχουμε
ως αποτέλεσμα

$$\vec{D}_2 = \epsilon_2 \vec{E}_2 = \epsilon_0 \left(3 - \frac{\rho}{c}\right) \left(\frac{3\kappa c}{\epsilon_0\rho(3c-\rho)}\right)\hat{\rho}_0 =$$
$$= \cancel{\epsilon_0} \left(\frac{3c-\rho}{\cancel{\rho}}\right) \frac{3\kappa\cancel{c}}{\cancel{\epsilon_0}\rho(3c-\rho)}\hat{\rho}_0 = \frac{3\kappa}{\rho}\hat{\rho}_0 \text{ (C/m}^2\text{)}$$

$$\ln\left(\frac{3c-\rho}{\rho}\right)' =$$
$$= \frac{1}{3c-\rho} \left(\frac{3c-\rho}{\rho}\right)' =$$
$$= \frac{1}{3c-\rho} \left(-\frac{3c}{\rho^2}\right) = \frac{-3c}{\rho(3c-\rho)}$$

(1): $a < \rho < c$ χρησιμοποιώντας Ν. Gauss

Είναι $\vec{D}$ το ίδιο
του Gaussiana

$$\oint_S \vec{D}_1 \cdot d\vec{S}_1 = Q_{\text{enclosed}} \rightarrow \int_0^{2\pi} \int_0^{2\pi} D_1 \hat{\rho}_0 \rho d\varphi dz \hat{\rho}_0 = Q \cdot \hat{\rho}_0$$

$$\rightarrow D_1 \cdot 2\pi \cdot \rho \cdot \cancel{\rho} = Q \cdot \cancel{\rho} \Rightarrow \vec{D}_1 = \frac{Q}{2\pi\rho} \hat{\rho}_0 \text{ (C/m}^2\text{)}$$

$$\vec{E}_1 = \frac{Q}{2\pi\epsilon_1\rho} \hat{\rho}_0 =$$
$$= \frac{Q}{4\pi\epsilon_0\rho} \hat{\rho}_0 \text{ (V/m)}$$

$$\varphi_1(\rho) = \int_{\rho}^{\infty} \vec{E}_1 \cdot d\vec{l} = \int_{\rho}^{\infty} \vec{E}_1 \cdot \hat{\rho}_0 d\rho + \int_{\infty}^{\infty} \vec{E}_2 \cdot d\vec{l}$$

$$\ll \int_p^c \frac{Q}{4\pi\epsilon_0 p} \hat{p}_0 \cancel{dp} \hat{p}_0 + \int_c^B \frac{3\pi c}{\epsilon_0 p(3c-p)} \hat{p}_0 dp \hat{p}_0 = \frac{Q}{4\pi\epsilon_0} \ln\left(\frac{c}{p}\right) + \frac{2\pi}{\epsilon_0} \ln 2, (v)$$

Διευκρινιστική σημείωση,

Δδν αντιστοιχεί συνθήκη $\rightarrow$

$$\varphi_2(c) = \frac{k}{\epsilon_0} \left(\ln 2 + \ln\left(\frac{3c-c}{c}\right) \right)$$

οκ ορίστηκε στο φερδ
αυφφ
για $p=c$

$$\varphi_2(p) = \frac{k}{\epsilon_0} \left(\ln 2 + \ln\left(\frac{3c-p}{p}\right) \right), \varphi_2(c)$$

$$\left| \begin{array}{l} p=c=2c \\ \varphi_2(c) \end{array} \right. = \frac{k}{\epsilon_0} \left(\ln 2 + \ln\left(\frac{3c-2c}{2c}\right) \right) = \frac{k}{\epsilon_0} (\ln 2 - \ln 2) = 0$$

$$\ln \frac{1}{2} = \ln 2^{-1} = -\ln 2$$

$$V_{A \rightarrow B} = \varphi_{1A} - \varphi_{1B} = \varphi_1(a) \ll \frac{Q}{4\pi\epsilon_0} \ln\left(\frac{c}{a}\right) + \frac{2\pi}{\epsilon_0} \ln 2, (v) \leadsto \text{η τιμή στα}$$

δύο του
συμμετά.

$$P_s|_{p=c} = n o_{1 \rightarrow 2} (D_2^{-1} - D_1^{-1})|_{p=c} \leadsto 0 = \hat{p}_0 \cdot \left(\frac{3\pi}{c} \hat{p}_0 - \frac{Q}{2\pi c} \hat{p}_0 \right) \Rightarrow \frac{3\pi}{c} = \frac{Q}{2\pi c}$$

$$\text{Άρα } V_{A \rightarrow B} = \frac{Q}{4\pi\epsilon_0} \ln\left(\frac{c}{a}\right) + \frac{2\pi Q}{3\epsilon_0} \ln 2$$

$$\ll \frac{Q}{V} = C = \frac{\ln(c/a) + \frac{\ln 2}{3\pi\epsilon_0}}{\frac{1}{4\pi\epsilon_0}} \rightarrow \left\{ k = \frac{Q}{6\pi} (c/m) \right\}$$

$$P.S|_{p=a} = \hat{\rho}_0 \cdot \left(\vec{D}_1 - \vec{D}_0 \right) \Big|_{p=a} = \hat{\rho}_0 \cdot \left(\frac{Q}{2\pi a} \hat{\rho}_0 \right) = \frac{Q}{2\pi a}, (C/m^2)$$

$$P.S|_{p=b} = \hat{\rho}_0 \cdot \left(\vec{D}_3 - \vec{D}_2 \right) \Big|_{p=b} = \hat{\rho}_0 \cdot \left(-\frac{3Q}{b} \hat{\rho}_0 \right) = -\frac{Q}{2\pi b}, (C/m^2)$$

$$(\gamma) \quad C' = \frac{1}{\frac{\ln(\frac{2a}{b})}{4\pi\epsilon_0} + \frac{\ln 2}{3\pi\epsilon_0}} = \frac{1}{\frac{\ln 2}{\pi\epsilon_0} \left(\frac{1}{4} + \frac{1}{3} \right)} = \frac{12\pi\epsilon_0}{7\ln 2}, (F/m)$$

$$\omega e' = \frac{1}{2} C' v^2 = \frac{1}{2} \frac{Q'^2}{C'} = \frac{1}{2} \frac{Q'^2 \cdot 7\ln 2}{12\pi\epsilon_0} = \frac{7Q'^2 \ln 2}{24\pi\epsilon_0}, (J/m)$$

$$\rightarrow \varphi^{(p)} = \frac{Q}{2\pi\epsilon} \ln(b/p), (V)$$

$$V = \varphi(a) - \varphi(b) = \frac{Q}{4\pi\epsilon_0} \ln(b/a) = \frac{Q}{4\pi\epsilon_0} \ln(4) = \frac{Q}{2\pi\epsilon_0} \ln(2) \rightarrow$$

$$\rightarrow C' = \frac{2\pi\epsilon_0}{\ln(2)}, (F/m), \quad \omega e' = \frac{1}{2} \frac{Q'^2}{C'}, (J/m)$$

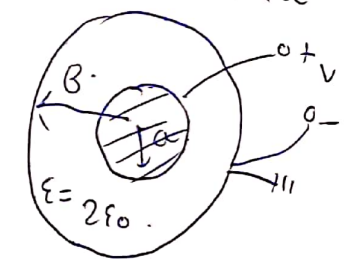
$\hookrightarrow C' \text{ ομογενής } C' \text{ ανισογενής } \sim \text{ μεταβολή χωρτιότητας, εμβαδού } n \text{ αλλαγών υλίου}$

$$(\gamma) \quad C = 2a, \\ C' (F/m), \\ \omega e' (J/m)$$

να αυξηθω b

αλλάζει η χωρτιότητα, $\epsilon = 2\epsilon_0$
 $\downarrow$
 στο $p=a, p=b$ να αυξηθ

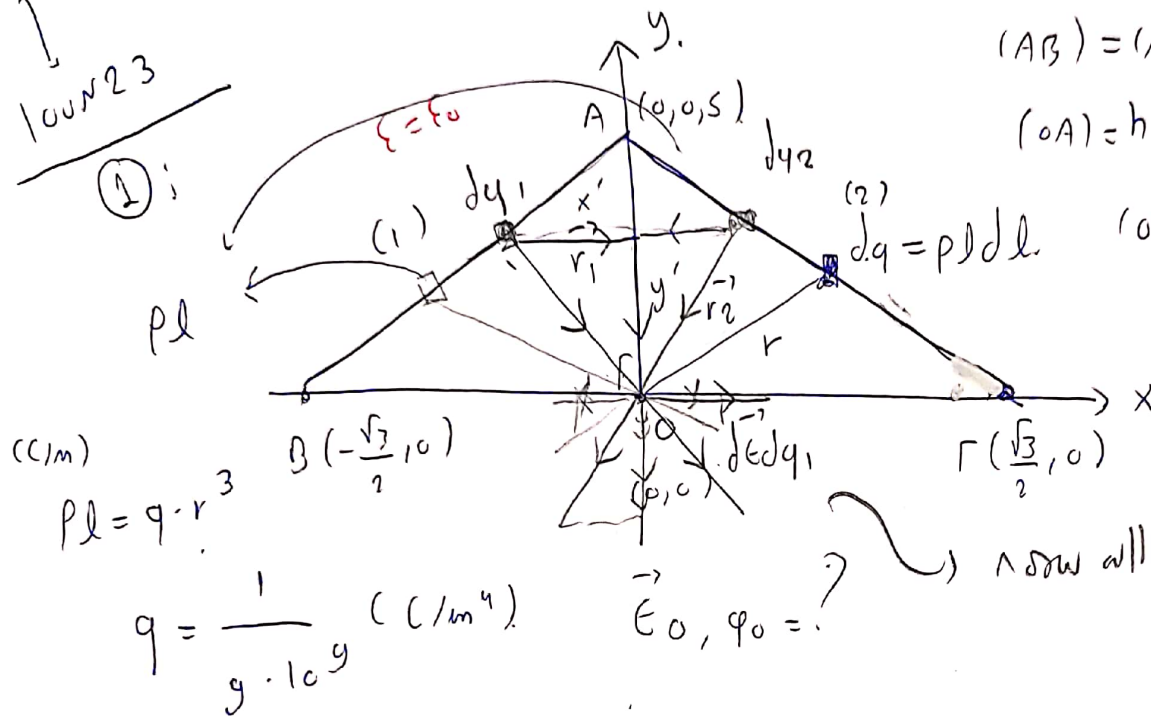
$$b = 2a = 2(2a) = 4a.$$



$$\rightarrow D = \frac{Q}{2\pi r} \hat{\rho}_0 \rightarrow$$

$$\rightarrow E = \frac{Q}{2\pi\epsilon r} \hat{\rho}_0$$

100N/23
①:



$$(AB) = (AG) = a = 1 \text{ m}$$

$$(OA) = h = 0.5 \text{ (m)}$$

$$(OB) = (OG) = \sqrt{(AB)^2 - (OA)^2} = \sqrt{1 - 0.25} = \sqrt{0.75} = \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}$$

(C/m)

$$\rho l = q \cdot r^3$$

$$q = \frac{1}{y \cdot \log y} \text{ (C/m}^4\text{)}$$

$$\vec{E}_0, \varphi_0 = ?$$

Αδωω αλληλεπιδράσεις με το $\vec{E}$ να μην είναι σαν $(-\hat{y}_0)$ γουστάρει.

$$d\vec{E}_0 = \frac{|dq|}{4\pi\epsilon_0 r_1^3} \vec{r}_1 = \frac{|\rho l| dl}{4\pi\epsilon_0 (x'^2 + y'^2)^{3/2}} [x' \hat{x}_0 + y' (-\hat{y}_0)]$$

$$\vec{r}_1 = x' \hat{x}_0 + y' (-\hat{y}_0) \quad \hat{x}_0 \perp \hat{y}_0$$

$$|\vec{r}_1| = \sqrt{x'^2 + (-y')^2} = \sqrt{x'^2 + y'^2}$$

$$r_1^3 = (x'^2 + y'^2)^{3/2}$$

$$\vec{r}_2 = x' (-\hat{x}_0) + y' (-\hat{y}_0)$$

$$r_2^3 = (x'^2 + y'^2)^{3/2}$$

Βρίσκω τον $\vec{E}$ λόγω (AG)

με το $\vec{E}$ της $AB \rightarrow$

$$y - y_B = \frac{y_A - y_B}{x_A - x_B} (x - x_B)$$

$$y - 0 = \frac{0.5 - 0}{0 - (+\frac{\sqrt{3}}{2})} (x + \frac{\sqrt{3}}{2})$$

$$y = \frac{1}{\sqrt{3}} (x + \frac{\sqrt{3}}{2})$$

$$y = -\frac{1}{\sqrt{3}} (x - \frac{\sqrt{3}}{2})$$

$$\int d\vec{E}_0 = \frac{q \cdot \cancel{r_1^2}}{4\pi\epsilon_0 (x'^2 + y'^2)^{3/2}} \left[x' \hat{x}_0 + y' (-\hat{y}_0) \right] =$$

$$= \frac{q \cdot \cancel{r_1^2}}{2\sqrt{3}\pi\epsilon_0} \left[x' \hat{x}_0 + y' (-\hat{y}_0) \right] =$$

$$= \frac{q}{2\sqrt{3}\pi\epsilon_0} \left[x' dx' \hat{x}_0 + \frac{1}{\sqrt{3}} \left(x' + \frac{\sqrt{3}}{2} \right) dx' (-\hat{y}_0) \right] \rightarrow$$

$$\rightarrow \vec{E}_0 = \frac{q}{2\sqrt{3}\pi\epsilon_0} \left[\int_{-\sqrt{3}/2}^0 \left(x' dx' \hat{x}_0 + \frac{1}{\sqrt{3}} \left(x' + \frac{\sqrt{3}}{2} \right) dx' (-\hat{y}_0) \right) \right] = \dots \rightarrow |d\vec{r}_1| = \sqrt{(dx')^2 + \left(\frac{1}{\sqrt{3}}\right)^2 (dx')^2} =$$

$$= \sqrt{1 + \frac{1}{3}} dx' = \frac{2}{\sqrt{3}} dx'$$

$$d\vec{E}_0 = \frac{1 dq_2}{4\pi\epsilon_0 r_2^2} \vec{r}_2 = \frac{\rho l d\vec{r}_2}{4\pi\epsilon_0 r_2^3} \vec{r}_2 =$$

$$= \frac{q \cdot \cancel{r_2^2}}{4\pi\epsilon_0 r_2^3} \frac{2}{\sqrt{3}} dx' \left[x' (-\hat{x}_0) + y' (-\hat{y}_0) \right] \quad d\vec{r}_2 = dx' \hat{x}_0 - \frac{1}{\sqrt{3}} dx' \hat{y}_0 \rightarrow$$

$$\downarrow \quad \rightarrow |d\vec{r}_2| = \frac{2}{\sqrt{3}} dx'$$

$$\hookrightarrow d\vec{r}_1 = dx' \hat{x}_0 + dy' \hat{y}_0, \text{ όπου}$$

$$y = \frac{1}{\sqrt{3}} x + \frac{1}{2} \rightarrow \frac{dy}{dx} = \frac{1}{\sqrt{3}} \rightarrow$$

$$\rightarrow dy = \frac{1}{\sqrt{3}} dx.$$

$$d\vec{r}_1 = dx' \hat{x}_0 + \frac{1}{\sqrt{3}} dx' \hat{y}_0 \rightarrow$$

$$d\vec{r}_2 = dx' \hat{x}_0 + dy' \hat{y}_0, \text{ όπου}$$

$$y = -\frac{1}{\sqrt{3}} \left(x - \frac{\sqrt{3}}{2} \right) = -\frac{1}{\sqrt{3}} x + \frac{1}{2} \rightarrow$$

$$\rightarrow \frac{dy}{dx} = -\frac{1}{\sqrt{3}} \rightarrow dy = -\frac{1}{\sqrt{3}} dx.$$

$$L, d\vec{t}_0 = \frac{q}{2\sqrt{3}\pi\epsilon_0} \left(x' d_{x'}(-\hat{x}_0) + \left(-\frac{1}{\sqrt{3}}x' + \frac{1}{2}\right) d_{x'}(-\hat{y}_0) \right) \rightarrow$$

$$\rightarrow \vec{t}_0 = \frac{q}{2\sqrt{3}\pi\epsilon_0} \left[\int_0^{\sqrt{3}/2} \underbrace{x' d_{x'}(-\hat{x}_0)}_{\left(\frac{x'}{2}\right)^2} + \int_0^{\sqrt{3}/2} \left(-\frac{1}{\sqrt{3}}x' + \frac{1}{2}\right) d_{x'}(-\hat{y}_0) \right] = \dots$$

$$d\varphi_0 = d\varphi dq_1 + d\varphi dq_2 = \frac{dq_1}{4\pi\epsilon r_1} + \frac{dq_2}{4\pi\epsilon r_2} = \frac{\rho l dl_1}{4\pi\epsilon r_1} + \frac{\rho l dl_2}{4\pi\epsilon r_2}$$

υπομνηστε
 να δώστε
 με την
 μέθοδο
 των ελαστικών
 για
 να δώτε την
 χρησιμοποιήστε